

Distinguishing learned performance from auditory bias: asymmetric perception of carryover and anticipatory nasal coarticulation

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Abstract

Nasal coarticulation occurs when a vowel (V) is (partly) nasalized adjacent to a nasal consonant (N) (Cohn 1993). Listeners can use nasal coarticulation as evidence for the presence of a nasal in an NV (Beddor & Onsuwan 2003) or VN (Beddor et al. 2018) sequence. However, VN (anticipatory) nasal coarticulation serves as evidence more often than NV (carryover) nasal coarticulation does, and it is more likely to interfere with existing nasal vowel-oral vowel contrasts (Delvaux et al. 2008) despite often being less prominent in articulation than carryover nasal coarticulation is (Jeong 2012). In a categorization task with bidirectional gating, English listeners used anticipatory but not carryover nasal coarticulation as evidence for adjacent nasal consonants. To explain this asymmetric response, two hypotheses were considered. First, carryover nasal coarticulation may be harder to detect because of a bias inherent to general auditory perception. Second, despite detecting both, listeners might disregard carryover but not anticipatory nasal coarticulation as evidence, because the latter is evidence for upcoming nasal consonants which generally speeds word recognition, while the former is evidence for previous nasal consonants which is only situationally relevant. In a 4IAX discrimination task, English listeners' perception of both carryover and anticipatory nasal coarticulation was shown to be veridical. While neither hypothesis was falsified, the selective adaptation account was supported, suggesting that the perceptual asymmetry arises primarily (but perhaps not exclusively) from language-specific knowledge which leads listeners to prioritize evidence from anticipatory nasal coarticulation and deprioritize evidence from carryover nasal coarticulation.

1 Introduction

Nasal coarticulation occurs when a vowel is partly or fully nasalized adjacent to a nasal consonant. Carryover nasal coarticulation occurs in NV sequences when the velum remains open after the nasal's oral constriction is released;

nasalization *carries over* from the preceding nasal to the vowel. Anticipatory nasal coarticulation occurs in VN sequences when the velum begins opening before the oral constriction is formed; the nasality of the following nasal is *anticipated* in the vowel (Cohn 1993). Just as listeners use coarticulation as evidence for how the speech stream should be segmented (Cutler 2012: ch. 6; Pisoni & Luce 1987), listeners may use nasal coarticulation as evidence for the presence of a nasal consonant with which that coarticulation is associated (Beddor & Onsuwan 2003, Ohala & Ohala 1995).

We might (naively) expect that in principle, whether nasal coarticulation can serve as a source of evidence for a nasal consonant should not be affected by whether that consonant precedes or follows the vowel. However, in practice, the evidence for nasal coarticulation having effects such as the neutralization of contrasts between oral and nasal vowels is much more abundant for anticipatory than for carryover nasal coarticulation (Jeong 2012: p. 448), suggesting that there may be an asymmetry between the perception of these two types of coarticulation.

Why should carryover and anticipatory nasal coarticulation have asymmetric perceptual effects? I consider two hypotheses. The first, which I refer to as the *auditory bias hypothesis*, is that the carryover and anticipatory nasal coarticulation are utilized as information sources asymmetrically because they are detected asymmetrically: sensitivity to nasality in vowels is reduced in NV sequences because the repeated presentation of a stimulus, in this case nasality, decreases sensitivity to that stimulus (Rusconi & Huber 2018). In other words, preceding nasals make nasal coarticulation harder to detect, but upcoming nasals don't. The second hypothesis, which I refer to as the *language-specific knowledge hypothesis*, is that listeners can detect both types of nasal coarticulation well enough to use them as sources of evidence about adjacent nasals, but that they weight evidence from anticipatory nasal coarticulation more heavily. In other words, the asymmetric treatment of carryover and anticipatory nasal coarticulation might reflect either an asymmetry in the *availability* of evidence to listeners, or an asymmetry in listeners' *utilization* of that evidence.

I hypothesize that i) an asymmetry between carryover and anticipatory nasal coarticulation exists at the level of listeners' weighting of evidence and is not inherent to articulation or to auditory perception, and ii) evidence for the identity of future segments is generally¹ useful for word recognition, but evidence for the identity of past segments is only situationally useful; listeners give greater weight (priority) to evidence which is more useful for word recognition, resulting in evidence from anticipatory nasal coarticulation being given greater weight and evidence from carryover nasal coarticulation being given less weight.

The paper is organized as follows. In §2, I present evidence from previous research that there is a perceptual asymmetry, and that the variation which exists

¹I do not say 'always' because predictive evidence only contributes to the determination of a word's identity up to that word's uniqueness point. For instance, by the time the anticipatory nasalized vowel [ẽĩ] is encountered in 'Mediterranean', the word's uniqueness point has already been reached; there is no alternative which also begins with [mɛ.rə.tə.r...]. Thanks to Amanda Rysling for pointing this out.

in the production of nasal coarticulation precludes an articulatory explanation and instead requires a perceptual one. In §3, I discuss an experiment which directly compares the perception of carryover and anticipatory nasal coarticulation in a categorization task, with the result that listeners utilize anticipatory but not carryover nasal coarticulation as evidence for adjacent nasal consonants. I argue that this is evidence for the existence of a perceptual asymmetry. In §4, I discuss a second experiment which also directly compares the perception of carryover and anticipatory nasal coarticulation, this time in a 4IAX discrimination task, with the result that perception of both coarticulations is veridical. I argue that this is evidence against the auditory bias hypothesis. In §5 I discuss psychophysical models of an auditory bias and language-specific weighting of evidence, how each might explain asymmetric perception, the empirical and typological predictions they make, and the necessity of directly testing the language-specific knowledge hypothesis. §6 is the conclusion.

2 Overview of nasal coarticulation

Coarticulation occurs when the articulation of one segment influences the articulation of an adjacent segment. Nasal coarticulation on vowels occurs when an otherwise oral vowel is (partially) nasalized adjacent to a nasal consonant. Oral vowels (V) are produced without any oral constriction and with a raised velum, permitting oral airflow but not nasal airflow, while nasal stops (N) are produced with a complete oral occlusion and a lowered velum, permitting nasal airflow but not oral airflow (Ladefoged & Maddieson 1996: pp. 103, 281). In an idealized NV or VN sequence, nasal airflow ceases just as oral airflow begins, or vice versa. When the two types of airflow overlap temporally, nasal coarticulation occurs.

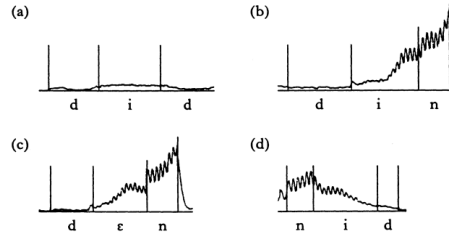


Figure 1: Nasal airflow traces for ‘deed’, ‘dean’, ‘den’, and ‘need’ show cline-like (increasing/decreasing) patterns because the velopharyngeal port opens/closes gradually through the duration of the vowel. Reproduced from Cohn (1993: Fig. 4).

In principle, carryover and anticipatory nasal coarticulation are both clines between fully oral and fully nasal states, and we might expect them to be identical apart from the direction of the cline, i.e., to be symmetric. In practice, these nasal coarticulations may be symmetric in some ways but not others, de-

pending on the language. For instance, in English, carryover and anticipatory nasal coarticulation appear to be roughly symmetric in temporal extent, but not in magnitude (as can be seen in Fig. 1). In general, the articulatory details of nasal coarticulation may be language-specific (Solé 1995, Pouplier et al. 2023) and even speaker-specific (Zellou 2022). Perception of nasal coarticulation may also be language-specific. The next two sections describe three studies which provide evidence that nasal coarticulation may serve listeners as a source of information about adjacent nasal consonants in a language-specific manner.

2.1 Carryover nasal coarticulation as evidence

Carryover nasal coarticulation serves as evidence that listeners use to resolve the contrast between /m/ and /^mb/ in Ikalanga (Bantu) (Beddor & Onsuwan 2003). The segment-internal acoustic attributes which serve as evidence for this contrast are relatively weak, i.e., it is difficult to distinguish /m/ from /^mb/ in isolation. However, /mV/ sequences have carryover nasal coarticulation on the vowel, while /^mbV/ sequences do not.

Beddor & Onsuwan presented Ikalanga listeners with a forced-choice categorization task in which participants categorized a stimulus word [gaCV] where C was either /m/ or /^mb/ and V was either oral [a] or carryover-nasalized [ã]. The possible responses were ‘gama’ and ‘gamba’. Responses varied primarily as a function of the presence or absence of carryover nasal coarticulation on the final /a/, rather than the presence or absence of a stop burst on /C/. In other words, the presence or absence of carryover nasal coarticulation is the primary type of evidence which listeners use to distinguish /m/ and /^mb/.

2.2 Anticipatory nasal coarticulation as evidence

Anticipatory nasal coarticulation serves as evidence for the presence of an upcoming nasal consonant in English. In a visual world eye-tracking experiment, Beddor et al. (2018) presented English listeners with images of /CVC/ and /CVNC/ minimal pairs such as ‘bed’ [bɛd] and ‘bend’ [bɛnd] (note the anticipatory nasal coarticulation), and played a recording of one of the words. After accounting for the time taken to plan and execute an eye movement, the listeners reliably began looking at the /CVNC/ words during the vowel, before the nasal consonant was audible. In other words, listeners distinguished /CVNC/ words from /CVC/ words as early as was possible based on anticipatory nasal coarticulation in the vowel.

English listeners also use anticipatory nasal coarticulation as evidence for upcoming nasals when those nasals are in a different syllable. In a forced-choice categorization experiment, Fowler & Brown (2000) presented English listeners with [CVCV] and [CVNV] tokens, some of which had vowels from /C_NV/ contexts cross-spliced into /C_CV/ and vice-versa, resulting in tokens in which the (lack of) anticipatory nasal coarticulation on the vowel was misleading with regards to whether the following consonant was nasal or oral. Listeners reported whether they heard ‘CVCV’ or ‘CVNV’, and were slower to respond to tokens

containing a misleading vowel, particularly when an oral vowel preceded a nasal consonant, showing that there was a processing cost associated with a mismatch between (lack of) anticipatory nasal coarticulation and the subsequent consonant.

Anticipatory nasal coarticulation, like any other potential cue, can only serve listeners as predictive evidence if it has high cue validity, i.e., if there is a strong correlation between the cue and what it signals. In a forced-choice categorization experiment, Ohala & Ohala (1995) presented English and Hindi listeners with gated recordings of /CVC/ and /CVN/ words in which the final consonant and sometimes part of the vowel was obscured with noise such that /CVC/ vs. /CVN/ became [CV?] vs. [C $\tilde{V}$?].

Both English and Hindi listeners discriminated between [CV?] and [C $\tilde{V}$?]. Given the possible responses ‘CVC’ and ‘CVN’, English listeners accurately categorized [CV?] as ‘CVC’ and [C $\tilde{V}$?] as ‘CVN’. Given the possible responses ‘CVC’, ‘CVN’, and ‘C $\tilde{V}$ C’, Hindi listeners accurately categorized [CV?] as ‘CVC’; however, [C $\tilde{V}$?] was categorized as ‘CVN’ or ‘C $\tilde{V}$ C’ with accuracy no better than chance.

In English, which does not have underlying nasal vowels, there is only one way for a nasal vowel to appear following an oral consonant: it must be part of a /VN/ sequence. In Hindi, which contrasts underlying oral and nasal vowels, there are two ways for a nasal vowel to appear following an oral consonant: it may be part of a /VN/ sequence and undergo anticipatory nasal coarticulation, or it may be an underlying nasal vowel / $\tilde{V}$ /. Thus, a [C $\tilde{V}$] sequence has high validity as a cue for a following nasal consonant in English, but not in Hindi.

2.3 Variation in coarticulatory patterns

There is strong evidence that the exact pattern of nasal coarticulation varies across languages. In a nasographic study comparing the relative timing of velar port opening with oral closure release in VN sequences, Solé (1995: pp. 8–9) finds that the opening of the velar port is timed relative to the duration of the vowel in English, but relative to the onset of the nasal’s oral closure in Spanish. In a nasal airflow study, Pouplier et al. (2023) find that anticipatory nasal coarticulation begins much earlier in the vowel in English compared to French and German.

There is also evidence of variation in nasal coarticulation within a single speech community. In a nasal airflow study, Zellou (2022) shows that English speakers at a single university campus could be divided into three groups who robustly showed positive, negative, or no correlation between the duration of nasal coarticulation in VN sequences and speech rate. In a study acoustically measuring the nasality of English vowels, Scarborough & Zellou (2013) show that the magnitude of anticipatory nasal coarticulation increases as the neighborhood density of a word increases.

These examples demonstrate that at least for anticipatory nasal coarticulation, it is possible for listeners to learn variation in coarticulatory patterns. While the presence of language-specific patterns of carryover coarticulation

might be taken as evidence against their non-veridical perception, the learning of language-specific coarticulation does not necessarily entail that listeners perceive it veridically or that it is encoded as a target or goal in the phonetics or phonology. It may be that even when coarticulation is perceived non-veridically, another factor which conditions the coarticulation *is* perceived veridically, and that is what is learned; the coarticulation emerges as a byproduct.

Given these possibilities, it is not safe to assume that systematicity in the production of a coarticulatory pattern entails its veridical perception or encoding of the pattern as a target in the phonetics or phonology. While listeners must learn something, the precise nature of what they learn is unclear. For this reason and for reasons which will be discussed in §5.3, I will make only very general and tentative claims about the relationship between the perception of nasal coarticulation and its production.

2.4 Interactions with contrast

In languages which contrast oral and nasal vowels, environments where nasal coarticulation occurs are often environments where this contrast is neutralized. For example, Ohala & Ohala (1995) demonstrated that Hindi listeners cannot distinguish between nasal vowels which are underlyingly specified as nasal and those which are the result of nasal coarticulation.

While there are examples of both carryover and anticipatory nasal coarticulation environments doubling as neutralizing environments, anticipatory nasal coarticulation more commonly results in neutralization. Strikingly, the neutralization of the /V $\tilde{V}$ / contrast in carryover contexts entails its neutralization in anticipatory contexts, but not vice versa (Jeong 2012: p. 448).

Some examples of languages in which the /V $\tilde{V}$ / contrast is neutralized are Hindi, Bengali, Lakota, and French. Hindi contrasts tautosyllabic [VC], [$\tilde{V}$ N], and [$\tilde{V}$ C]; there is no [VN] to contrast with [$\tilde{V}$ N] (Ohala & Ohala 1995: p. 44); in other words, (tautosyllabic) anticipatory nasal coarticulation neutralizes the /V $\tilde{V}$ / contrast. The facts are the same in Bengali (Lahiri & Marslen-Wilson 1991: pp. 259–260). In Lakota, the /V $\tilde{V}$ / contrast is neutralized immediately following a nasal consonant. Lakota is not a clear example of neutralization by carryover nasal coarticulation, however; out of six speakers, three neutralized the contrast after nasals, while three maintained the contrast (Scarborough et al. 2015: p. 300).

French, which contrasts [VC], [VN], and [$\tilde{V}$ C] (Cohn 1993: p. 51), is a notable case of neutralization for several reasons. Anticipatory nasal coarticulation ([$\tilde{V}$ N]) is present, but only in high vowels, where there is no /V $\tilde{V}$ / contrast. The /V $\tilde{V}$ / contrast is still neutralized before tautosyllabic nasals, but to oral [V] rather than nasal [$\tilde{V}$]. Carryover nasal coarticulation is robust but does not neutralize the /V $\tilde{V}$ / contrast (Delvaux et al. 2008).

2.5 Direct comparisons of the perception of carryover and anticipatory nasal coarticulation

Carryover and anticipatory nasal coarticulation pattern differently in production and in perception. In production, in terms of temporal extent and amount of nasalization (measured acoustically or by nasal airflow), carryover nasal coarticulation can be at least as prevalent as anticipatory nasal coarticulation, sometimes more so (Cohn 1993, Delvaux et al. 2008). But in perception, anticipatory nasal coarticulation appears more prevalent. It is more likely than carryover nasal coarticulation to be realized as local categorical assimilation, and more likely to neutralize /V $\tilde{V}$ / contrasts (Jeong 2012: p. 448).

In the current literature, there is more evidence for listeners using anticipatory nasal coarticulation as a source of evidence than there is for listeners using carryover nasal coarticulation the same way. However, there has not been a systematic cross-linguistic survey of the perception of carryover and anticipatory nasal coarticulation, and the studies which do exist have been inconsistent in controlling for factors such as the contrasts which exist within the language under investigation and whether nasal coarticulation originates from a tautosyllabic nasal consonant, making direct comparisons between the perception of carryover and anticipatory nasal coarticulation difficult. To the best of my knowledge, the only previous work which attempts to directly compare the perception of carryover and anticipatory nasal coarticulation is Jeong (2012). I will discuss this study in some detail because it illustrates a mode of argumentation which I have attempted to avoid in this work.

A typological survey reveals what Jeong describes as a disparity between carryover and anticipatory nasal coarticulation in phonetics and phonology: cross-linguistically, carryover nasal coarticulation is more temporally extensive than anticipatory nasal coarticulation, but anticipatory nasal coarticulation is far more likely to be a categorical (and thus, Jeong claims, phonological) process of local assimilation. The explanation posited for this disparity is that onsets, being more perceptually salient than codas, trigger more complete parsing of coarticulation out of adjacent vowels. Thus, nasalization is more fully parsed out of the vowel following a nasal onset than preceding a nasal coda, making carryover nasal coarticulation more difficult to perceive and thus less likely to be phonologized than anticipatory nasal coarticulation.

Jeong supports this account with an AXB discrimination experiment comparing the perceived nasality of vowels in carryover and anticipatory nasal coarticulation contexts. The vowels were drawn from a four-step synthetic continuum between fully oral and fully nasal. The X vowel was presented either in a carryover nasal coarticulation frame [am_] or an anticipatory nasal coarticulation frame [b_m]. 21 native Korean listeners were more likely to judge vowels as similar to the less nasal of A and B when they were presented in the carryover frame as opposed to the anticipatory frame. Jeong interprets this result to mean that vowels are perceived as less nasal than they actually are in carryover but not anticipatory nasal coarticulation contexts, consistent with her hypothesis.

Though ambitious in its aim of explaining an apparent paradox of phono-

logical and phonetic typology, this approach encounters several problems. First, AXB tasks are unable to disentangle listeners’ discrimination abilities from their language-specific categorization behavior² (Gerrits & Schouten 2004), meaning that results from this type of task cannot safely be used to argue for generalizations about cross-linguistic perception. Second, it explicitly predicts that non-veridical perception of nasal coarticulation should be conditioned by the syllable role of the associated nasal consonant. Anticipatory nasal coarticulation in heterosyllabic $V]_{\sigma}N$ sequences should thus pattern with carryover nasal coarticulation in being perceived non-veridically, but according to the results of the typological survey, this is not the case (Jeong 2012: p. 449).

Finally, and most seriously, the mode of argumentation in this paper relies on a flawed premise: it implicitly assumes that the only factors affecting the perception of nasal coarticulation are factors related to production (i.e., gestural parsing), and consequently, it is safe to make claims about the treatment of nasal coarticulation in the production-oriented phonetics and phonology based on assumptions about how it is perceived. The possible contributions of factors such as language-specific evidence weighting or biases inherent to general audition are not considered. Setting aside the fact that theories of gestural parsing have encountered serious challenges (Diehl, Lotto & Holt 2004), we do not yet have a principled theory of the relationship between phonology and the psychophysical factors affecting perception, meaning that it is not safe to assume that we can make testable predictions about one based on the other.

The present study does not attempt to make definitive claims about the effects that asymmetric perception of carryover and anticipatory nasal coarticulation might have on phonetic or phonological typology. While there is discussion of the possible effects in §5.3, it is speculative. What is attempted here is a direct comparison of the perception of carryover and anticipatory nasal coarticulation to understand whether there is in fact an asymmetry, and if so, why that asymmetry should exist.

3 Experiment 1

The goal of Experiment 1 was to determine whether English listeners display a perceptual asymmetry by testing whether they use carryover and anticipatory nasal coarticulation symmetrically as sources of evidence about adjacent nasal consonants. It can be considered an attempted replication of Ohala & Ohala (1995) comparing the responses to two types of nasal coarticulation in one language rather than responses to one type in two languages. It can also be considered an attempted replication of Beddor & Onsuwan’s (2003) finding that carryover nasal coarticulation is a source of evidence for onset nasals.

Throughout this section, I use $/_N\alpha/$ $/\alpha_N/$ to represent the vowel $/\alpha/$ with carryover and anticipatory nasal coarticulation, respectively.

²Jeong (2012: p. 456) reasons that because Korean does not have significant nasal coarticulation, listeners’ language-specific knowledge should not affect the results. No citation or additional argumentation is given.

3.1 Methods

The experiment consisted of a categorization task in which participants heard a CVC, CVN, or NVC syllable with either the first consonant (for CVC vs. NVC) or last consonant (for CVC vs. CVN) and sometimes part of the vowel gated off and replaced with white noise, leaving the cline of nasality on the vowel (none, decreasing, or increasing) as the only source of evidence for the identity of the deleted consonant. This syllable was always one of ‘bob’ (CVC), ‘mob’ (NVC), or ‘bomb’ (CVN). Participants were given a forced choice between responding with ‘bob’ or ‘mob’ (when gating from the left edge), or ‘bob’ or ‘bomb’ (when gating from the right edge). It involved the manipulation of three independent variables: the direction of gating (left edge gated vs. right edge gated); the nasality of the stimulus (oral ‘bob’ vs. nasal ‘mob’/‘bomb’); and gate length (50 ms, 100 ms, 150 ms, 200 ms, and the full syllable).

Given that English listeners had previously been shown to utilize anticipatory nasal coarticulation to discriminate between /CVC/ and /CVN/ syllables, the anticipatory (‘bob’ vs. ‘bomb’) condition was treated as a control, with the expected result being successful discrimination despite the the final consonants being gated off, i.e., replication of the English results from Ohala & Ohala (1995). Failure to discriminate in the anticipatory condition would indicate that the results in the carryover condition were not meaningful.

Two hypotheses were considered regarding the results of the carryover condition (‘bob’ vs. ‘mob’). The null hypothesis was that participants are unable to discriminate between ‘bob’ and ‘mob’ when the initial consonant has been gated off because they are insensitive to the difference between oral vowels and vowels with carryover nasal coarticulation.

The research hypothesis was that participants are able to discriminate between ‘bob’ and ‘mob’ when the initial consonant has been gated off because they can detect carryover nasal coarticulation and use it as a source of information about the preceding consonant, analogous to their use of anticipatory nasal coarticulation as a source of information about the upcoming consonant.

A null result (failure to discriminate between ‘bob’ and ‘mob’) would not be informative about whether asymmetric perception of carryover and anticipatory nasal coarticulation is attributable to an auditory bias or to language-specific knowledge, since both would explain the result equally well. However, if participants successfully discriminated between ‘bob’ and ‘mob’ using carryover nasal coarticulation, it would be evidence against the auditory bias hypothesis, because listeners would have to perceive carryover nasal coarticulation well enough that they could use it as a source of evidence for the missing preceding nasal.

3.1.1 Participants

11 participants were recruited by word of mouth. They were not compensated for their time, and were informed of this fact before the experiment began; participation was thus on a volunteer basis. Ages ranged from 20 to 65 (mean 31.36, median 25). All were native listeners of English. One participant’s results

were discarded due to a reported childhood hearing disorder; the remaining ten participants reported no history of hearing or neurological disorders.

3.1.2 Materials

Each item was one of the syllables /bab/ ‘bob’, /mab/ ‘mob’, or /bam/ ‘bomb’, with one consonant and part of the vowel gated off and replaced with white noise. White noise was appended to the gated tokens to avoid a sharp transition to silence being interpreted as a stop, a possible confound in Lahiri & Marslen-Wilson (1991) pointed out by Ohala & Ohala (1995). White noise was chosen because it is a non-linguistic sound, and participants were explicitly informed that they would hear words obscured with noise, both factors encouraging participants to interpret the white noise as noise overlaid on a word rather than part of the word itself.

All audio manipulation and analysis was done in Praat (Boersma & Weenink 2023). A male native speaker of English recorded 10 instances of ‘bob’, ‘mob’, ‘bomb’, and ‘mom’ in the frame ‘Guess ____ soon.’ The recordings were performed in a quiet sound booth using a GLS Audio ES-57 microphone connected via a Focusrite Scarlett Solo (3rd Gen) to a PC running Praat. The recordings were low-pass filtered at 10 kHz and denoised with the “reduce noise” function on default settings. All instances of /b/, /m/, and /a/ were segmented manually; boundaries between consonants and /a/ were identified by the beginning or end of /a/’s steady-state formants (i.e., before the visible beginning or end of formant transitions), while boundaries between the consonants and the frame sentence were identified by the beginning or end of noise from /s/.

The tokens were constructed by splicing together productions of /b/, /m/, and /a/ from recordings of the syllables ‘bob’, ‘mob’, and ‘bomb’. The vowel of ‘bob’ was always oral. The vowel of ‘mob’ had a falling cline of nasality (carryover nasal coarticulation); the vowel of ‘bomb’ had a rising cline of nasality (anticipatory nasal coarticulation).

Seven segment types were identified for extraction from the recordings. These were onset and coda /b/, onset and coda /m/, /a/, /_Na/, and /a_N/. The first instance of each segment which met certain requirements was extracted. Selected /m/s had to have modal voicing with a duration of at least 90 ms. Selected /a/s had to have modal voicing, a falling pitch contour, and a duration of at least 200 ms. Care was taken to ensure that the different vowels had similar pitch contours to lessen the chance that listeners would rely on the pitch contour to distinguish different tokens. The vowels were cropped to 200 ms, taking the middle portion of the vowel (at most 20 ms of material was removed in total in each case). There were no special requirements for /b/.

The /b/s were taken from productions of fully oral ‘bob’ and the /m/s from fully nasal ‘mom’ in order to ensure that the segments were completely oral or nasal without traces of a transition between those states. The /b/s were approximately 60 ms in duration, onset /m/ 100 ms, and coda /m/ 120 ms.

These segments were spliced together to create three tokens: /bab/, /m_Nab/, and /ba_Nm/. These were then gated. /m_Nab/ was gated from the left edge,

/ba_Nm/ from the right, and /bab/ from both sides, producing two series of gated /bab/ tokens. The gates, in milliseconds of vowel remaining, were: 50, 100, 150, 200 (full vowel minus consonant), and the full vowel plus consonant. The excised material was replaced with 200 ms of white noise generated in Praat. All resulting stimuli were scaled to an average intensity of 70 dB.

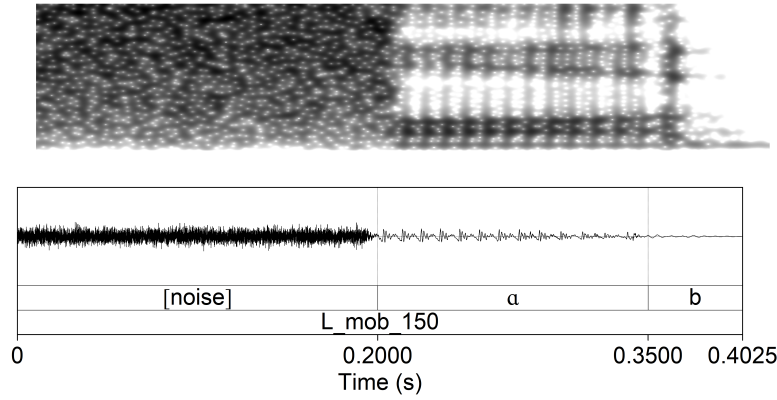


Figure 2: A spectrogram and waveform of ‘mob’ gated from the left edge with 150 ms of the vowel remaining. The gated material has been replaced with 200 ms of white noise.

3.1.3 Nasality of the stimuli

In order to verify that the stimulus vowels showed the expected cline-like nasality, a script (Styler 2017a) was used to take measurements in a specified number of windows equally spaced throughout the each vowel. Repeated measurements showed that the first and last windows produced unreliable and unstable values, so these were discarded. The primary measure used was A1–P0 (compensated), a reliable measure of vowel nasality in English (Styler 2017b) which measures the difference between A1 (the amplitude of the highest harmonic in F1) and P0 (the amplitude of the highest of the first or second harmonic, corresponding to a nasal pole), with adjustments based on nearby formants which reduce the effect of vowel quality (Chen 1997). Lower values of A1–P0 correspond to greater nasality.

Figure 3 shows average values of A1–P0 (compensated) across all recordings of ‘bob’, ‘mob’, ‘bomb’, and ‘mom’. The average ‘mom’ vowel is highly nasalized, while the average ‘bob’ vowel is not. The clines of nasality in ‘bomb’ and ‘mob’ are also visible. The average ‘bomb’ vowel displays a clear transition from orality comparable to the ‘bob’ vowel to nasality comparable to the ‘mom’ vowel by its midpoint. The cline of nasality in the average ‘mob’ vowel is less dramatic, but still visible, beginning near the ‘mom’ vowel and ending near the

‘bob’ vowel.

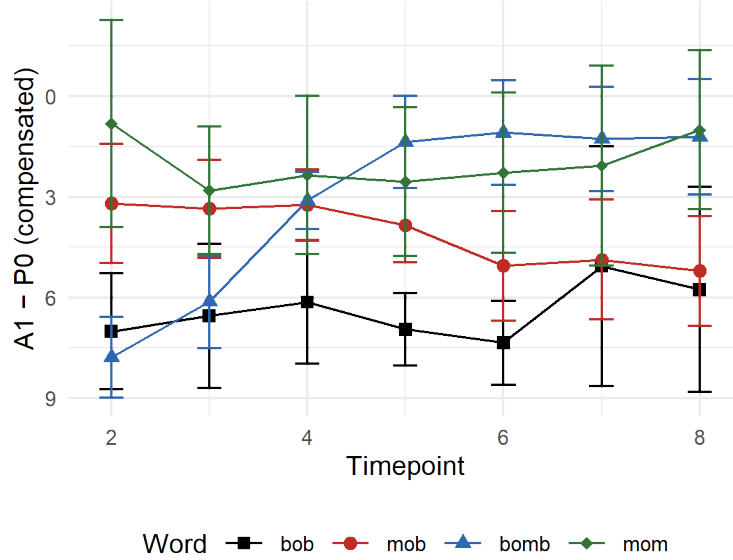


Figure 3: Average A1–P0 (compensated) at timepoints 2 through 7 of vowels in CVC (‘bob’), NVC (‘mob’), CVN (‘bomb’), and NVN (‘mom’) syllables, with error bars of 1 standard deviation. Smaller values correspond to greater nasality. Note the inverted y-axis.

In addition to acoustic measurements of nasality, three phonetically trained native listeners of English listened to isolated vowels extracted from ‘bob’, ‘mob’, and ‘bomb’. Their impressionistic judgements were that the ‘bob’ vowel seemed fully oral, the ‘mob’ vowel had a decreasing cline of nasality, and the ‘bomb’ vowel had an increasing cline of nasality.

3.2 Procedure

The experiment was conducted online via PCIBex (Zehr & Schwarz 2018). Participants completed a consent form and filled out a brief questionnaire about their demographic information, language experience, and any past hearing or neurological disorders. They then underwent a headphone check to ensure that they were wearing headphones.

The experimental task was forced-choice categorization. Each trial was preceded by a 1000 ms pause, during which the screen was blank. In each trial, a stimulus token played as two written words appeared on screen. The words were either “bob” and “mob” or “bob” and “bomb”. Each word appeared over a large “F” or “J”; “bob” always appeared over “F”; “mob” and “bomb” appeared over “J”. The participants responded by pressing either the F or J key

on their keyboard to report which word they believed they heard. The trial ended immediately if a key press was recorded, or after 2000 ms if there was no response. The experimental procedure, including participant compensation and consent, were approved by the UC Santa Cruz Internal Review Board.



Figure 4: A screen capture of a trial in Experiment 1.

The experiment consisted of a 20-trial warm-up, followed by four 80-trial blocks separated by short breaks—320 experimental trials, and 340 trials in total. Excluding the warm-up, all items appeared 16 times each.

After the task was completed, participants filled out a questionnaire about their experience in the experiment itself. Two participants reported that the stimuli became inaudible partway through the experiment; this is a known failure mode of PCIBex (Emily Knick, p.c.). These participants reported that they did not respond to these trials, and an inspection of their recorded responses showed that they did cease responding approximately halfway through. A statistical analysis excluding these participants was not substantially different from the original analysis; consequently, the final analysis was conducted with data from all ten participants.

3.3 Results

A proportion correct analysis showed three main results. First, performance in the oral (‘bob’) conditions was similar for left- and right-edge gating, as expected given that there should not be a substantial difference in the correlations of the onset and coda /b/s. Second, performance increased as gate length increased, reaching a maximum of roughly 70%. This was also expected, and conforms with the intuition that the more time the listener has to accumulate evidence, the more accurate their judgment should be.

Third, and most strikingly, performance in the nasal (‘mob’/‘bomb’) conditions was markedly differentiated by direction of gating and nasal coarticulation. In the right-edge/anticipatory nasal (‘bomb’) condition, performance was at roughly 70% correct at all gates except the 50 ms gate, showing that English listeners are able to detect anticipatory nasal coarticulation and use it as evidence for the nasality of the upcoming segment having heard just the first half of the vowel. In the left-edge/carryover nasal (‘mob’) condition, performance was at chance at every gate below 200 ms, while in the right-edge/anticipatory

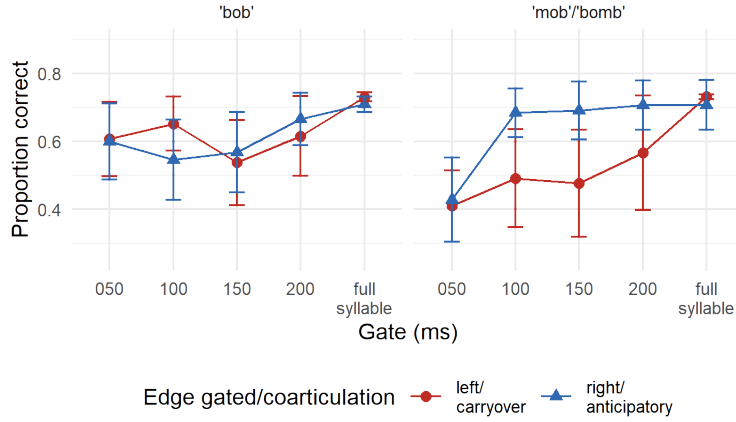


Figure 5: Proportion correct at gates of 50–200 ms and with the full syllable in each cross of the oral (‘bob’, left) vs. nasal (‘mob’/‘bomb’, right) and left-edge gating/carryover nasal coarticulation (‘bob’/‘mob’, red) vs. right-edge gating/anticipatory nasal coarticulation (‘bob’/‘bomb’, blue). Error bars represent one standard deviation.

nasal (‘bomb’) condition performance was at chance at the 50 ms gate and at roughly 70% correct from the 100 ms gate onwards.

3.4 Discussion

The high performance observed with right-edge gating, particularly in the nasal condition, may be interpreted as constituting a successful replication of the English portion of Ohala & Ohala (1995), who also found a dramatic increase in proportion correct at longer gates. The low performance observed with left-edge gating, however, cannot be considered a successful replication either of a time-reversed Ohala & Ohala (1995) or Beddor & Onsuwan (2003). In other words, listeners’ ability to categorize syllables based on their perception of nasal coarticulation was decidedly asymmetric, strongly favoring anticipatory nasal coarticulation. There are three hypotheses which might account for this result.

The first and least interesting is that listeners’ performance was asymmetric because the stimuli were asymmetric: the ‘mob’ vowel showed a shallower cline of nasality and was less nasal overall compared to the ‘bomb’ vowel (see Fig. 3). In other words, the difference between the ‘mob’ and ‘bob’ vowels was smaller than the difference between the ‘bomb’ and ‘bob’ vowels, making it harder to distinguish gated stimuli in the carryover condition than in the anticipatory condition. This hypothesis could be tested by rerunning Experiment 1 with symmetric stimuli, such as those described in §4.1.2. Obtaining the same results would falsify it; obtaining different results (e.g., equal performance in the carryover and anticipatory conditions) would confirm it.

The other two hypotheses are of course the auditory bias and language-specific knowledge hypotheses. The former would explain the asymmetric results as the consequence of asymmetric perception: evidence for onset /m/ was simply not as readily available to listeners as evidence from coda /m/ because carryover nasal coarticulation was perceived non-veridically (or at least more non-veridically than anticipatory nasal coarticulation). In the context of this experiment, this possibility cannot be distinguished from the asymmetric stimulus hypothesis described above. The language-specific knowledge hypothesis would explain the results as the consequence of English listeners heavily weighting evidence from anticipatory nasal coarticulation while disregarding evidence from carryover nasal coarticulation, and being unable to adjust the weighting of evidence from carryover nasal coarticulation in time to make a difference in their performance. This may be unexpected from the perspective of neurological adaptation to changing distributions of sounds, which typically takes roughly 60 seconds (Simpson et al. 2014), but it is not clear whether whatever mechanism governs language-specific evidence weighting works by the same rules.

Indeed, neurological adaptation cannot be the only mechanism behind language-specific evidence weighting. Listeners may adjust category boundaries to accommodate different speech rates, non-native accents, or even the idiomatic productions of individual speakers, while remaining unable to perceive non-native sound contrasts despite considerable exposure (Norris, McQueen & Cutler 2003), and it is not obvious whether the ability to use nasal coarticulation as a source of evidence should behave more like adaptation to speech rate or adaptation to a non-native contrast. Even if neurological adaptation is the mechanism at work, and we should expect adaptation after roughly 60 seconds, it is not an argument for the auditory bias hypothesis: the nasality of the ‘mob’ vowels may simply have been too subtle to detect until the 200 ms gate.

Experiment 1 was not designed to distinguish between the auditory bias and language-specific knowledge hypotheses. What it successfully does is demonstrate that English listeners, when presented with stimuli which are representative of those they encounter in everyday speech (i.e., stimuli with asymmetric clines of nasal coarticulation), exhibit asymmetric utilization of anticipatory and carryover nasal coarticulation as evidence. More importantly, placing English listeners in a task which demands that they make categorization judgments based on the presence or absence of carryover nasal coarticulation is not informative about how they perceive nasal coarticulation. An experiment which tests one of the above hypotheses to the exclusion of the others is necessary. Experiment 2 is designed to accomplish this exclusion insofar as is possible.

4 Experiment 2

Experiment 1 shows that English listeners will utilize anticipatory but not carryover nasal coarticulation as a source of evidence about adjacent nasal consonants. However, it does not directly address the research question: What are the relative contributions of auditory perception and language-specific knowledge to

the carryover-anticipatory coarticulation asymmetry?

One strategy for approaching an answer to this question is to test the hypotheses separately in sequence, starting with the auditory bias hypothesis. However, separating the contributions of the auditory system and language-specific knowledge so that they may be tested separately is non-trivial. Many researchers have observed that speech perception and language-specific knowledge influence each other. From a phonological perspective, language-specific phonological knowledge guides speech perception while also being shaped by it (Beddor et al. 2018, Zellou 2022), and it is not obvious how to separate the two in an experiment.

However, from a psychophysical perspective, the auditory bias hypothesis is essentially the claim that the failure to perceive carryover nasal coarticulation veridically is a failure of discrimination: listeners are unable to discriminate between [NV] and [N $\tilde{V}$]. If listeners cannot discriminate two classes of stimuli, then they cannot categorize them differently. Testing the auditory bias hypothesis thus requires testing discrimination independently of categorization. (This is why the results of Experiment 1 cannot be used to evaluate the auditory bias hypothesis: Experiment 1 was a categorization task.)

Categorization is, of course, language-specific. For example, the voice onset time at which listeners switch between categorizing stops as voiced or voiceless is language-specific (Keating, Mikoś & Ganong 1981). However, categorization is also task- and context-specific. Category boundaries may shift depending on the adjacent phonetic context (Holt & Lotto 2010: p. 1222), or depending on the range of stimuli presented (Keating, Mikoś & Ganong 1981). Listeners rely on more than one acoustic attribute to identify each individual sound, and a decrease in one attribute opposed by an increase in another may result in the same percept, e.g., the perception of stops as voiced or voiceless depends not only on VOT but the ratio between VOT and aspiration noise amplitude (Repp 1979). “Categorical perception” itself is likely an artifact of categorization tasks, which include both same-different judgments in a laboratory setting and the rapid series of categorization judgments listeners must perform in order to recognize phonemes and/or words in everyday speech (Holt & Lotto 2010). The time course of the task also affects how categorical listeners’ judgments are: stimuli which are available to listeners for longer are less likely to be perceived categorically. For example, Pisoni (1973) found that within-category discrimination is at chance for stops, higher for 50 ms vowels, and highest for 300 ms vowels.

This last example can be understood in terms of information density, or the ratio between the amount of information in the stimulus and how long the listener is exposed to it. Assuming that there is a limit on how quickly listeners can uptake information from the speech signal and that greater information uptake leads to better discrimination performance, results like those of Pisoni (1973) are expected due to the greater information density of stops compared to vowels.³ For a given stimulus, the longer a listener is exposed to it, the

³Stops have much greater information density than vowels for two reasons. First, stops present the listener with more information than vowels do: in addition to formant structure

longer they have to uptake more detailed information about it; the more detailed information they have, the more fine-grained their judgments are likely to be. In other words, the continuation of a stimulus over a longer duration spreads out its information over time, decreasing its information density, and the information density of the stimulus is inversely correlated with the likelihood that judgement of that stimulus will be categorical.

It is this final property, information density, which is relevant to Experiment 2. Experiment 2 uses a 4-interval AX (4IAX) paradigm. In a traditional AX paradigm, participants report whether two stimuli, A and X, were the same or different; in a 4IAX paradigm participants hear two pairs of stimuli, and report which pair contained a difference. For example, given an item (set of four stimuli) AB-AA, the correct responses is “first different”; given AA-AB, the correct responses is “second different”. Unlike 2-interval paradigms, which have a tendency to bias participants towards the use of stimulus-external category labels (i.e., towards categorization), judgments in 4-interval paradigms can be made with reference to stimulus-internal qualities only, and tend to bias participants to focus on the acoustic qualities of the stimuli (Gerrits & Schouten 2004: p. 375). The 4-interval paradigm gives listeners three chances to hear one of the stimuli, decreasing information density and thus the likelihood of categorical behavior.

4.1 Methods

The experiment consisted of a 4IAX forced-choice discrimination task in which participants heard two pairs of synthetic vowels with nasal coarticulation drawn from a 10-step oral-nasal continuum and indicated which pair contained the odd vowel out. It involved the manipulation of three independent variables of interest: the base step (out of the bottom six of the continuum) of each stimulus set; the size of the difference between stimuli (1 or 4 continuum steps); and the direction of nasal coarticulation (carryover vs. anticipatory).

This experiment was designed to test the auditory bias hypothesis by observing whether perception of carryover nasal coarticulation is inhibited even when the influence of language-specific knowledge is reduced, i.e., when auditory factors should be the primary ones determining performance. There are two general patterns of results which would bear on this question.

The first pattern would be different performance in the carryover vs. anticipatory nasal coarticulation conditions. Assuming that the 4IAX paradigm successfully minimizes the influence of language-specific knowledge, then greater performance in the anticipatory condition indicates that the perceptual asymmetry may be attributable to a bias inherent to auditory perception rather than language-specific knowledge.

The second pattern would be similar performance in the carryover and anticipatory conditions. This would demonstrate that listeners are able to distinguish

in the form of formant transitions, they also have release bursts. Second, stops present their information over a much shorter duration than vowels do.

different levels of carryover nasal coarticulation either despite or without the influence of an auditory bias, suggesting that the perceptual asymmetry cannot originate from auditory perception and must originate elsewhere. In particular, similar performance in the carryover and anticipatory conditions indicates that the perceptual asymmetry must be attributed to language-specific knowledge.

The critical manipulation is thus direction of nasal coarticulation. The other manipulations of interest, size of difference and base step, serve to provide control conditions: performance in the 1-step difference condition is expected to be poorer than in the 4-step condition or even at chance based on the assumption that such small differences in nasality will be very difficult for listeners to detect, and performance is not expected to vary substantially across the six base steps.

4.1.1 Participants

30 participants (ages 18–31, mean 21.30, mode 20) completed the experiment. All but one participant were native American English listeners. They were recruited from the UC Santa Cruz undergraduate participant pool via the UC Santa Cruz Department of Linguistics’ online Linguistics Experiments Participant Sign-Up system or via word-of-mouth and compensated with either course credit or a payment of \$12.

4.1.2 Materials

The stimuli were syllables spliced together from a combination of recordings and synthesized vowels with various clines of nasality. The vowels were produced via a Klatt synthesizer running in Praat; this synthesizer accepts specifications for the frequency of formants over a time interval and returns a synthesized audio file. The specifications were based on time-normalized averages of recordings of the author’s vowels in ‘bob’ and ‘bomb’. The synthetic vowels were 250 ms in length.

Two continua were generated via interpolation between two endpoints, one with carryover nasal coarticulation and one with anticipatory nasal coarticulation. The endpoints of the anticipatory continuum were the average ‘bob’ and ‘bomb’ vowels; the endpoints of the carryover continuum were the average ‘bob’ vowel and a reversed version of the average ‘bomb’ vowel. A reversed ‘bomb’ vowel was used instead of a ‘mob’ vowel to ensure that the vowels would be identical except for the direction of the nasality cline, i.e., that the stimuli would be symmetric, a property which the stimuli of Experiment 1 lacked. Recordings of /b/ taken from a ‘bob’ recording were spliced onto the beginning and end of each synthetic vowel, resulting in a final stimulus shape of /bVb/ which was 421 ms in length. In future iterations of this experiment, formant frequency and bandwidth will be manipulated, as these have the greatest effects on the perception of nasality in vowels (Styler 2015: p. 162).

Each item contained four stimuli. The manipulations were: i) which of the four stimuli was the odd one out; ii) the direction of difference (higher or lower nasality than the majority); iii) the size of the difference (1 vs. 4 continuum

steps); and iv) the direction of nasal coarticulation in the stimuli (all carryover vs. all anticipatory). This resulted in a total of 32 different items for each continuum step. The manipulations of interest were (iii) difference size and (iv) direction of nasal coarticulation. The other two manipulations were not taken into account as factors in the analysis; collapsing them together results in eight trials for each cross of factors (iii) and (iv).

Given a 10-step continuum, there were six possible 4-step differences: 1-5, 2-6, 3-7, 4-8, 5-9, and 6-10. In order to keep the number of 1-step and 4-step items the same, steps 1 through 6 were also taken as the base steps of the 1-step differences as well. With six base steps and 32 items per step, the total number of trials (before repetition) would be 192. In order to decrease the number of trials, an additional manipulation was introduced, odd vs. even base steps, and treated as a between-participant manipulation. Half of the participants' items had continuum steps 1, 3, and 5 as their base step; the other half's items had continuum steps 2, 4, and 6 as their base step. Consequently, a full block of trials consisted of $3 \text{ steps} \times 32 \text{ trials} = 96 \text{ trials}$.

4.2 Procedure

The experiment was conducted in a quiet room on a desktop computer running PsychoPy (Peirce et al. 2019). Participants completed a consent form and read an instruction sheet before beginning the experiment. They heard an item over headphones and reported whether the first or second pair contained a difference by pressing the “F” or “J” keys (to indicate a difference in the first or second pair, respectively) on the computer keyboard. The experiment consisted of 5 practice trials of increasing difficulty, followed by 32 acclimation trials, and finally 384 experimental trials (4 repetitions $\times$ 96 trials per block), for a total of 421 trials. The acclimation trials appeared identical to the experimental trials to participants, but the results were recorded separately in order to increase the chance that participants' performance would stabilize by the time the experimental trials were reached.

The time course of each trial was as follows. At the beginning of each trial, a fixation cross appeared in the center of the screen. The first stimulus played from 0 to 432 ms, the second from 500 to 932 ms, the third from 1250 to 1682 ms, and the fourth from 1750 to 2182 ms; thus, the within-pair inter-stimulus interval was 68 ms, and between-pair ISI was 318 ms. The fixation cross disappeared at 2360 ms (178 ms after the end of the final stimulus) and was replaced by a large ‘F’ and ‘J’. Participants had 2000 ms to respond before the experiment automatically advanced to the next trial.

The participants were given breaks of up to 5 minutes after the acclimation trials and every 96 trials (approximately every 5–7 minutes) after that. After the experiment, participants completed questionnaires about their demographics and the experiment itself, and read a debriefing document explaining the purpose of the experiment. Including time to complete forms, the experiment lasted between 30 and 45 minutes. The experimental procedure, including participant compensation and consent, were approved by the UC Santa Cruz

Internal Review Board.

A formal calculation of the experiment’s power was not conducted. However, as an alternative, a goal of at least 20 observations per condition per participant (OPCPP) was used as a rough guideline. Increasing OPCPP reduces the contribution of each individual trial to the final result, lessening the chance that non-experimental factors which alter the results of individual trials (such as a participant momentarily losing focus) will introduce noise into the data. With eight trials per condition per block and four blocks per participant, the total OPCPP was $8 \text{ trials per condition} \times 4 \text{ repetitions} = 32$, more than 150% of the target number.

4.3 Results

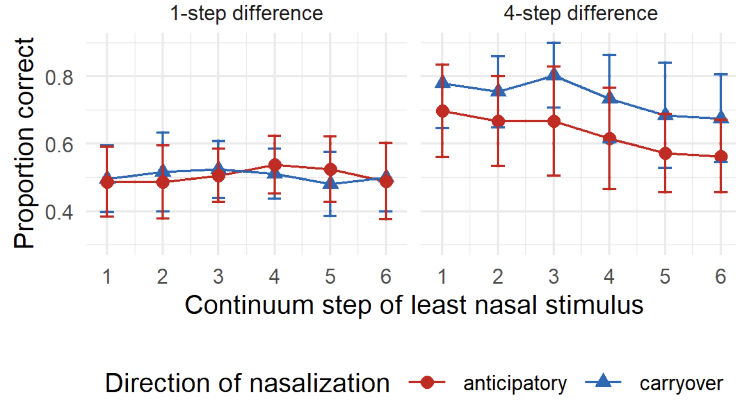


Figure 6: Proportion of responses correct in the 1-step difference (left) and 4-step difference (right) conditions. Error bars represent one standard deviation.

Results were measured by taking proportion correct (pc) of responses (a detection theory analysis is forthcoming). Trials in which a response was not recorded were not included in the analysis. In the 1-step difference condition, performance was indistinguishable from chance with both carryover and anticipatory nasal coarticulation. In the 4-step difference condition, performance was above chance in both the carryover and anticipatory conditions, and decreased as base step increased while remaining above chance; performance in the carryover condition was somewhat higher at all base steps. The primary factor affecting performance was the size of the difference between stimuli, not direction of nasal coarticulation.

4.4 Discussion

The results did not show the pattern which would be expected (greater performance in the anticipatory condition) if there were an auditory bias which resulted in the non-veridical perception of carryover nasal coarticulation. Indeed, while direction of nasal coarticulation had an effect in the 4-step difference condition, the effect was the opposite from what would be expected based on the results of Experiment 1: English listeners performed better in the 4-step $\times$ carryover condition than the 4-step $\times$ anticipatory condition, even though Experiment 1 established that English listeners use anticipatory but not carryover nasal coarticulation as evidence for adjacent nasal consonants. In other words, English listeners can discriminate different levels of carryover nasal coarticulation at least as well as anticipatory nasal coarticulation, but only use the latter coarticulation as a source of evidence.

These results are compatible with the hypothesis that listeners preferentially attend to predictive information (i.e., anticipatory nasal coarticulation): while English listeners can perceive both kinds of coarticulation, they have no reason to use carryover nasal coarticulation as a source of evidence, but they do have reason (the acceleration of word recognition) to use anticipatory nasal coarticulation as a source of evidence for upcoming nasal consonants.

To sum up, the asymmetric perception of carryover and anticipatory nasal coarticulation is not attributable to a bias inherent to auditory perception, but may be attributable to speakers utilizing the different coarticulations asymmetrically as sources of evidence.

5 General discussion

So far, in discussing the auditory bias and language-specific knowledge hypotheses, I have left both relatively underspecified. The two experiments presented here have served to provide evidence that a perceptual bias against carryover nasal coarticulation exists, and that this bias disappears in a task which tests discrimination rather than categorization. Though the difference between these results is striking, it is not clear whether they are sufficient to fully evaluate either hypothesis. In order to do so, we must posit specific theories of how an auditory bias and language-specific knowledge might be instantiated, and determine whether the experimental results falsify or support either of them. I begin by describing a plausible mechanism which could result in an auditory bias against the veridical perception of carryover nasal coarticulation: neural habituation.

5.1 The auditory bias hypothesis as habituation

The perceptual wink model (Rusconi & Huber 2018) is a model of perception in which the various attributes of a stimulus may be detected separately by distinct ‘detectors’. For instance, the perception of a letter such as “E” may be decomposed into the detection of its identity (‘fifth letter of the Latin alphabet’)

and its case ('upper-case') by a separate letter identity detector and letter case detector. The authors position the perceptual wink model as an extension of the nROUSE (neural responding optimally with unknown sources of evidence) model of priming. The nROUSE model assumes that negative priming effects are the result of neural habituation to a stimulus, which prevents the undesirable blending of sequential activations, but also results in failure to register rapid repetitions of a stimulus (Huber & O'Reilly 2003).

A brief diversion into terminology is necessary. Physically, a detector is a synapse or set of synapses, and its activation is those synapses firing in response to a particular stimulus. The habituation of a detector represents the post-firing fatigue of the corresponding synapses. The accumulation of sufficient activation will cause the synapses to fire again, whether this activation accumulates gradually or all at once. For this reason, I will use the term 'accumulator' rather than Rusconi & Huber's (2018) 'detector'.

In the perceptual wink model, when an accumulator encounters its stimulus, activation peaks in the first 100–200 ms, before falling as habituation begins around 200–400 ms. Reactivating the accumulator during this habituation period is difficult; consequently, if additional stimuli are presented during the habituation period, the accumulator may require a longer period of uninterrupted exposure to build up activation gradually, or may simply fail to activate entirely. Crucially, multiple accumulators relevant to a particular stimulus may not only operate independently, but undergo habituation and fail independently (hence 'perceptual wink'—one eye is open, but the other is closed). For instance, an observer may successfully perceive a letter's identity but not its case, or vice versa, as Rusconi & Huber (2018) demonstrate in a pair of rapid serial visual presentation experiments.

To implement an auditory bias against the perception of carryover nasal coarticulation using the perceptual wink model, we must minimally posit at least one nasality accumulator and some means of measuring the relative duration of nasality. For a listener to perceive that nasal coarticulation has occurred (that nasalization is present both on a nasal consonant and on the vowel adjacent to it), they must perceive that the nasalization extends over both consonant and vowel.

From the viewpoint of the recognition of (sequences of) sounds and coarticulation affecting them, failure to perceive nasalization on the vowel means that nasal coarticulation is unavailable to the listener as a source of evidence for or about the nasal consonant; failure to perceive nasalization on the consonant may result in an $[N\tilde{V}]$ or $[\tilde{V}N]$ sequence being perceived as $[C\tilde{V}]$ or $[\tilde{V}C]$, respectively; and failure to perceive the consonant entirely may result in an $[N\tilde{V}]$ or $[\tilde{V}N]$ sequence being perceived as lone $[\tilde{V}]$.

From the viewpoint of the recognition of words, the duration of nasality relative to the segments of the word may be a useful or even crucial difference which distinguishes words. For instance, words with higher neighborhood densities are more difficult to recognize due to many similar-sounding words simultaneously being activated, but for English listeners, exaggerated nasal coarticulation ameliorates these effects (Scarborough & Zellou 2013). To an Ikalanga listener, the

difference between ‘gama’ and ‘gamba’ depends mostly on whether nasality extends into the post-nasal vowel or not (Beddor & Onsuwan 2003). Failure to perceive the relative duration of nasality veridically (or at least well enough for it to serve as a source of evidence) may result in slowdowns in word recognition like those observed by Fowler & Brown (2000) or even in failures to discriminate between similar words.

It is possible at this point to determine that the results of neither Experiment 1 nor Experiment 2 can be used to evaluate the perceptual wink account of an auditory bias. The perceptual wink account makes predictions about the perception of sequences of nasalized sounds; the gated nasal condition stimuli in Experiment 1 were of the shape $[C\tilde{V} ?]$ and $[? \tilde{V} C]$ (where ? represents the gated consonant), and the stimuli in Experiment 2 were of the shape $[b\tilde{V} b]$. These stimuli lacked nasal consonants which might cause habituation of the nasality accumulator(s), so a habituation-based auditory bias would have no detectable effects.

In principle, it might still be possible for habituation to be detected in Experiment 2: if the first vowel in a stimulus pair triggered habituation, the second vowel might be perceived non-veridically as different, resulting in participants incorrectly reporting a difference in that pair. Indeed, the first stimulus pair was 1.30 times more likely to be misidentified as containing a difference than the second pair, and many participants reported that they did not bother listening to the second stimulus pair if they perceived a difference in the first. However, this may simply reflect a more general bias towards choosing the first pair, which was 1.23 times more likely to be chosen than the second pair regardless of whether it was the correct response. Additionally, the time course of habituation does not match the ISIs. As stated earlier, habituation occurs 200–400 ms after activation, but the second stimulus did not begin playing until 500 ms into each trial.

In order to properly evaluate the perceptual wink account of the auditory bias hypothesis, Experiment 2 must be rerun with stimuli of the shape $[N\tilde{V}N]$. The ISIs might also be adjusted to preclude habituation from one item affecting the perception of another. These modifications remain to be realized. For now, I will continue describing the perceptual wink model’s predictions with regards to the perception of nasal coarticulation.

In $[N\tilde{V}]$ and $[\tilde{V}N]$ sequences, the listener is faced with the problem of whether to attribute nasality to the consonant, the vowel, or both. A simplistic solution to this problem is to take the relative duration of the nasality accumulator’s activation at face value and attribute nasality to any segment which coincided with the nasality accumulator being activated (analogous to the manner in which associations are formed between letter identities and upper or lower case).

Thus, in an $[N\tilde{V}]$ sequence the nasality accumulator is fully activated by the N but becomes habituated during the $\tilde{V}$, resulting in non-veridical perception of the $\tilde{V}$ ’s nasality.

This may strike some readers as reminiscent of compensation for coarticulation (Ohala et al. 1981), and the proposed net effects for $[N\tilde{V}]$ sequences are similar, but compensation and habituation should be understood as fundamentally

different processes. Compensation for coarticulation posits that non-veridical perception is the result of listeners mentally reversing coarticulation to create a non-coarticulated percept which was not physically present in the speech signal. The listener is aware that distortion has occurred in articulation, and mentally corrects it. The habituation account posits that non-veridical perception is the result of contextually reduced sensitivity to attributes of the speech signal. The distortion that occurs is not in articulation, but in the listener’s auditory perception, and the listener takes the distorted signal at face value. Compensation for coarticulation also assumes that the task of the listener is to parse the speech signal into the series of articulatory gestures which produced it; the habituation account makes no such assumption.

In any event, the perceptual wink model makes a different prediction for $[\tilde{V}N]$ sequences. In a $[\tilde{V}N]$ sequence the nasality accumulator will be activated by the $\tilde{V}$. A lightly nasalized vowel might prime the nasality accumulator without fully activating it, allowing it to respond more quickly to following N; this prediction is consistent with Fowler & Brown’s (2000) cross-splicing results. A heavily nasalized vowel might fully activate the nasality accumulator, resulting in the percept of a nasal vowel and potentially in habituation to nasality which results in non-veridical perception of the following N’s nasality. An additional prediction is that because anticipatory nasal coarticulation is lighter before onset nasals than before coda nasals Krakow 1999: p. 36, there should be priming before onset nasals but habituation before coda nasals.

To sum up, a sufficiently nasalized segment will activate the nasality accumulator, resulting in the perception of that segment as nasal, but also in non-veridical perception of the following segment’s nasality. This predicts non-veridical perception of $\tilde{V}$ in $[N\tilde{V}]$ and N in $[\tilde{V}N]$. The latter prediction is unproblematic for languages like English and French, in which the (non-)nasality of the next segment is predictable from the nasality of the vowel. However, it is a problem for languages like Hindi, in which the nasality of the post- $\tilde{V}$ segment is unpredictable.

Evidently, there must be some way to surmount this difficulty, because languages like Hindi do exist. I cannot provide any definitive answers as to how this happens, but I can provide some educated speculation. One possibility is that rather than a single nasality accumulator, there is a set of accumulators which each respond to some acoustic correlate of nasality, e.g., a nasal zero accumulator, a nasal pole accumulator, an increased-formant-bandwidth accumulator, etc. whose outputs are combined to yield an overall “nasal” percept. The existence of trading relations (Repp 1979, 1982, Parker, Diehl & Kluender 1986) shows that the percept of properties such as voicing is the aggregate of multiple, sometimes many acoustic attributes of the speech signal; Styler (2015: pp. 165–166) shows that A1 – P0, P0 prominence, and formant bandwidth all contribute to the perception of nasality in vowels, suggesting that perceived nasality may also be an aggregate of multiple accumulators’ output. $\tilde{V}$ and N might activate different subsets of these accumulators, resulting in veridical perception of both segments.

Alternatively, non-veridical perception of $[N\tilde{V}]$ but veridical perception of

$[\tilde{V}N]$ would occur if the accumulators that $\tilde{V}$ activates are a proper subset of those that N activates; in $[N\tilde{V}]$, N would habituate all nasality accumulators, resulting in non-veridical perception of $\tilde{V}$, but in $[\tilde{V}N]$, $\tilde{V}$ would leave one or more nasality accumulators untouched and ready to activate for N .

The relative importance (i.e., weighting) of the various nasality accumulators might differ cross-linguistically. The productions of nasal consonants and vowels might also differ cross-linguistically, perhaps with the effect that there is a subset-superset relation between $\tilde{V}$'s accumulators and N 's accumulators (and a consequent bias against the veridical perception of carryover nasal coarticulation) in some languages but not others. These articulations might also be adjusted to better activate specific accumulators. For instance, there is evidence that French nasal vowels are articulated with breathy voice so as to exaggerate their already large nasal spectral tilt (Styler 2017b: p. 2481); in the perceptual wink model, this would be the for the benefit of a spectral tilt accumulator.

An additional prediction of the perceptual wink model is a dispreference for short sequences of nasal sounds. Such sequences could be perceived non-veridically due to habituation from the first nasal segment making it more difficult to perceive that nasality has continued through more than one segment. Meanwhile, singleton nasal sounds would not be affected by habituation, and long sequences of nasal sounds would be able to overcome habituation by building up activation in the nasality accumulator(s) gradually even after habituation has occurred. In phonological terms, we might expect either singleton nasal sounds with no nasal spreading, or unbounded nasal spreading producing long sequences of nasal sounds. It is not clear exactly how many segments would constitute a “long” or “short” sequence. The hypothesized time course of habituation is 200–400 ms after initial activation (Rusconi & Huber 2018), enough to cover one or several segments depending on their duration, which varies considerably (Klatt 1976: p. 1210).

The perceptual wink account of an auditory bias against the perception of carryover nasal coarticulation makes a variety of testable predictions. The most readily testable of these is that a version of Experiment 2 run with $[N\tilde{V}N]$ stimuli should see a decrease in discrimination performance in the carryover condition.

5.2 The language-specific knowledge hypothesis as selective adaptation

An approach which has the potential to model language-specific perceptual knowledge of nasal coarticulation is selective adaptation by the human auditory system. Simpson et al. (2014) argue for a mechanism for selective adaptation which is sensitive to i) the probability distributions of sounds and ii) the relative similarity of sounds. This mechanism adjusts sensitivity to sounds based on both of these factors. Sensitivity to a low-probability sound is adjusted downward if that sound is relatively similar to high-probability sounds, and upward if it is dissimilar (i.e., an “oddball”); sensitivity to high-probability sounds remains average. These sensitivity changes take place after roughly one minute of evidence-gathering (i.e., listening).

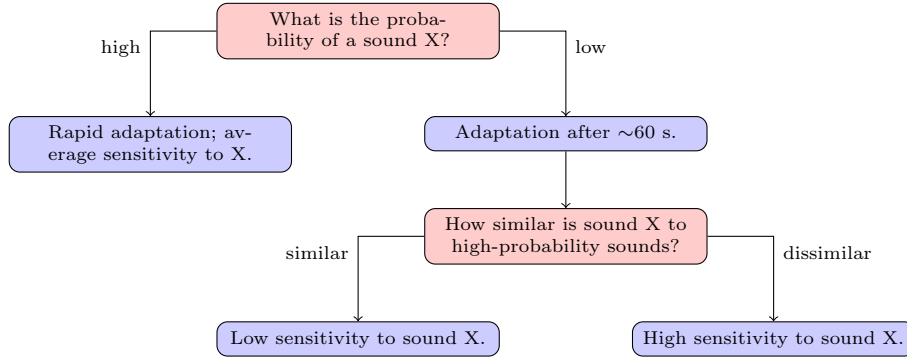


Figure 7: A flowchart summarizing the outcomes generated by the statistical learning mechanism proposed in Simpson et al. (2014).

The authors demonstrate this in a pair of AX tasks in which participants heard pairs of wideband noise bursts with slight variations added to their base intensity and reported which was louder. These bursts were drawn from a set of three sounds which varied either in base intensity or ISI and which had skewed probabilities of occurring in each trial: one sound had an 80% probability of occurring, while the other two each had a 10% probability of occurring. They found that discrimination performance was predictable from similarity between the sounds: if the high-probability sound was at one extreme of the stimulus set (lowest/highest intensity or shortest/longest ISI), then sensitivity to the low-probability sound at the opposite extreme (the oddball) increased while sensitivity to the intermediate low-probability sound decreased, and the divergence in sensitivity occurred after roughly one minute of exposure to the stimuli. If the high-probability sound was intermediate between the low-probability sounds, no divergence occurred: neither low-probability sound was an oddball. Adaptation was also continuous throughout the experiments: the probability distribution of the three sounds was changed every 100 trials, and each time, participants adapted after approximately one minute.

The selective adaptation account of auditory perception is capable of explaining some observations in both experiments (including an otherwise mysterious observation in Experiment 2). In Experiment 1, there were twice as many ‘bob’ trials as there were ‘bomb’ or ‘mob’ trials, so the bob vowel [a] was a high-probability sound while the ‘mob’ vowel [N a] and the ‘bomb’ vowel [a N] were lower-probability sounds. The nasality of the ‘mob’ vowel was intermediate between that of ‘bob’ and ‘bomb’; measuring by nasality, the ‘bomb’ vowel was the most dissimilar to the ‘bob’ vowel, and thus was the oddball. The selective adaptation account predicts that sensitivity to the high-probability ‘bob’ vowel should be average, sensitivity to the oddball ‘bomb’ vowel should be high, and sensitivity to the low-probability but insufficiently distinctive ‘mob’ vowel should be low. This is precisely the result obtained in Experiment 1: sensitivity to the ‘mob’ vowel was low, sensitivity to the ‘bomb’ vowel was high, and

sensitivity to the ‘bob’ vowel was intermediate between them.

This implies that the results of Experiment 1 were about the overall level of nasality in the vowels rather than the direction of nasal coarticulation. A consequent prediction is that if Experiment 1 were rerun with nasal stimuli that had equivalent levels of nasality, the difference in performance between the carryover nasal and anticipatory nasal conditions should disappear. Further, because neither nasalized vowel would be the oddball and attract unusually high sensitivity, sensitivity in the nasal conditions should not be any higher than in the oral condition, and perhaps lower.

In Experiment 2, unlike in Experiment 1, there were two continua comprising nineteen total stimuli (the most oral step of both continua was the same). However, we can think of these stimuli as being more or less distant from three broad prototypes roughly equivalent to those in Experiment 1: oral, nasal with carryover coarticulation, and nasal with anticipatory coarticulation. Stimuli in steps 1–6 of the continua appeared as both the low and high nasality stimuli in items, while steps 7–10 only appeared as the high steps, meaning that relatively oral stimuli were more common than relatively nasal stimuli. If one type of low-probability relatively nasal stimuli were perceived as more dissimilar from the relatively oral stimuli, then that type would become the oddball (or set of oddballs), and sensitivity to it would increase, while sensitivity to the other type decreased.

This may explain two otherwise puzzling effects observed in the results of Experiment 2. These effects are i) greater performance in the 4-step difference carryover condition than in the 4-step difference anticipatory condition, and ii) decreasing performance at higher (more nasal) base continuum steps (e.g., in the 4-step anticipatory condition, proportion correct is nearly 15% lower at step 6 than at step 1). The difference between the carryover and anticipatory conditions could result from the carryover stimuli being perceived as more dissimilar from the oral baseline than the anticipatory stimuli were. Admittedly, this is difficult to quantify, and would involve similarity comparisons between a much larger number of stimuli than were used to demonstrate selective adaptation in Simpson et al. (2014).

A somewhat less speculative possibility is that the second effect, decreasing performance at more nasal base steps, could be explained by steps 7–10 being less common than steps 1–6. If steps 7–10 were perceived as roughly equally dissimilar to the higher probability steps (i.e., if none of steps 7–10 was an oddball), then sensitivity to them should be below average. In a phenomenological model of the influence of stimulus probabilities on neurons’ adaptational states over a series of trials, when the low-probability stimuli were equally similar to the high-probability stimulus (none was the oddball), sensitivity to the high-probability stimulus was predicted to be average, while sensitivity to the low-probability stimuli was predicted to be low (Simpson et al. 2014: figure 4). The rolloff in performance at higher base steps in the 4-step difference condition could thus be explained as the higher-step stimuli having a lower probability of occurrence.

While the results of experiments 1 and 2 may thus tentatively be interpreted

as supporting the selective adaptation account, the same issue raised in the previous section—the lack of $[\tilde{N}\tilde{V}]$ and $[\tilde{V}\tilde{N}]$ sequences in the stimuli—again limits how well we may evaluate this account. In the selective adaptation approach, sensitivity to a stimulus is a function of its probability and relative similarity to high-probability stimuli. However, the probability that a particular stimulus will occur is in turn a function not only of its commonness in the overall set of stimuli, but also its context. The likelihood of a given sound occurring in a certain preceding context is that sequence’s transitional probability.

Let us consider the transitional probabilities (TPs) of $[\tilde{V}\tilde{N}]$, $[\tilde{N}\tilde{V}]$, $[\tilde{N}\tilde{V}]$, and $[\tilde{N}\tilde{V}]$. The TP of $[\tilde{V}\tilde{N}]$ and $[\tilde{N}\tilde{V}]$ varies from language to language. In English, the TP of $[\tilde{V}\tilde{N}]$ is high while the TP of $[\tilde{N}\tilde{V}]$ is low due to robust anticipatory nasal coarticulation. In Hindi, the TP of $[\tilde{V}\tilde{N}]$ is low due to anticipatory nasal coarticulation, but the TP of $[\tilde{N}\tilde{V}]$ is middling because $\tilde{V}$ might be followed by N or C . The results of Ohala & Ohala (1995) can be thought of in terms of TP: English listeners consistently labeled gated $[\tilde{C}\tilde{V}]$ as $/\tilde{C}\tilde{V}\tilde{N}/$ because the TP of $[\tilde{V}\tilde{N}]$ is high in English, but Hindi listeners responded with both $/\tilde{C}\tilde{V}\tilde{N}/$ and $/\tilde{C}\tilde{V}\tilde{C}/$ because the transitional probabilities of $[\tilde{V}\tilde{N}]$ and $[\tilde{V}\tilde{C}]$ are both non-zero in Hindi. In French, the TP of $[\tilde{V}\tilde{N}]$ is high and the TP of $[\tilde{N}\tilde{V}]$ is low in high vowels, which show anticipatory nasal coarticulation; the situation is reversed in non-high vowels, where anticipatory nasal coarticulation is absent.

Meanwhile, the TPs of $[\tilde{N}\tilde{V}]$ and $[\tilde{N}\tilde{V}]$ sequences, however, show no apparent cross-linguistic variation. The cross-linguistic TP of $[\tilde{N}\tilde{V}]$ appears to be zero; $/\tilde{N}\tilde{V}/$ is apparently always realized with carryover nasal coarticulation (Jeong 2012: p. 445, Stanton 2016: p. 1092, and references therein). However, there are still $/\tilde{N}\tilde{V}\tilde{N}\tilde{V}/$ contrasts, e.g., French *mot* $/m\tilde{o}/$ ‘word’ vs. *mon* $/m\tilde{o}/$ ‘my (masculine)’ (Delvaux et al. 2008: p. 602). Clearly, listeners must be sensitive to the temporal extent of nasalization as well as its magnitude. Simpson et al. (2014) observe selective adaptation to variations in ISI, showing that the adaptational mechanism is sensitive to variations in the time course of stimuli. How this is accomplished is left underspecified, but for the purposes of selective adaptation, it appears that we may treat lexical nasal vowels as distinct from vowels with carryover nasal coarticulation.

There is also the case of languages like Ikalanga. While $[\tilde{N}]$ and $[\tilde{N}\tilde{C}]$ may be articulatorily distinct, telling them apart in perception without reference to external sources of evidence is difficult, and many languages which permit an $/\tilde{N}\tilde{N}\tilde{C}/$ contrast before vowels prohibit the same contrast word-finally, i.e., in positions where there is no carryover nasal coarticulation to serve as an external source of evidence for the contrast, but not vice versa (Stanton 2016: p. 1098). In other words, in such languages the $/\tilde{N}\tilde{V}\tilde{N}\tilde{C}/$ contrast ends up behaving as if it were a $[\tilde{N}\tilde{V}\tilde{N}\tilde{V}]$ contrast. Consequently, I proceed with the assumption that the TPs of $[\tilde{N}\tilde{V}]$ and $[\tilde{N}\tilde{V}]$ in languages like Ikalanga can be treated as functionally equivalent to those of $[\tilde{N}\tilde{V}]$ and $[\tilde{N}\tilde{V}]$. Beddor & Onsuwan’s (2003) finding that the perception of the $/ma\ m\tilde{b}a/$ contrast depends primarily on carryover nasal coarticulation, and that the presence or absence of the stop burst makes very little difference, supports this view.

Importantly, not all $/\tilde{N}\tilde{N}\tilde{C}/$ contrasts will depend on carryover nasal coartic-

ulation. For instance, in English, the /N NC/ contrast is permitted in word-final position (e.g., ‘ban’ vs. ‘band’), suggesting that the presence or absence of a stop burst is sufficient to support the contrast. Additionally, nasal onsets and codas both condition anticipatory nasal coarticulation, but the coarticulation conditioned by onsets is of lesser magnitude than that conditioned by codas Krakow 1999: p. 36, which could distinguish, for instance, ‘candy’ (with heavy nasalization) from ‘canny’ (with light nasalization) if the burst proved insufficient. Carryover nasal coarticulation is consistently present after [N] and absent after [NC], but it is unlikely to be the primary source of evidence for the contrast, unlike in Ikalanga.

We can now approach a theory of language-specific sensitivity to carryover nasal coarticulation. To recapitulate, in the selective adaptation approach, sensitivity to a stimulus is a function of i) its probability of occurring, and ii) its similarity to other high-probability stimuli. The probability of a nasalized vowel occurring depends both on context and on contrast: in a language that does not contrast oral and nasal vowels, the occurrence of nasalized vowels depends entirely on context; in a language that *does* contrast oral and nasal vowels, the occurrence of nasal vowels depends only partly on context.

In languages such as English, $\tilde{V}$ occurs in two contexts: N_- , and ${}_N$. (The French facts are the same for high vowels; for low vowels, (lexical) low $\tilde{V}$ occurs in the contexts N_- and ${}_C$ rather than N_- and ${}_N$.) Following contexts are not considered in calculations of TP; we must instead compare preceding contexts, i.e., the probability of $\tilde{V}$ in N_- vs. in C_- , $\#_-$, V_- , etc. For simplicity, let us only consider the TPs of $[N\tilde{V}]$ and $[C\tilde{V}]$. The transitional probability of $[N\tilde{V}]$ is high, i.e., $\tilde{V}$ is a high-probability sound in the context N_- , and sensitivity to it in that context is thus predicted to be average. The TP of $[C\tilde{V}]$ is lower; more precisely, it is the probability of the sequence $/CVN/$. In the context C_- , if the probability of $\tilde{V}$ is lower than the probability of V , and if $\tilde{V}$ is sufficiently dissimilar from V , then $\tilde{V}$ is an oddball and sensitivity to it in that context is predicted to be high. The results of Experiment 1 may reflect this, with discrimination performance between $[ba?]$ and $b[\tilde{a}]$ being at roughly 70% correct at all but the 50 ms gate.

In languages such as Hindi, (carryover) $\tilde{V}$ occurs with high probability in N_- , and anticipatory and lexical $\tilde{V}$ occur in C_- presumably with lower probabilities than V . The sensitivity facts should thus be similar to those in English. A potential difference from English is the level of sensitivity to N in $[\tilde{V}N]$. An issue which was raised in the discussion of the perceptual wink model was that it may predict reduced sensitivity to N in $[\tilde{V}N]$ due to habituation of the nasality accumulator(s), but Hindi contrasts $[\tilde{V}C]$ and $[\tilde{V}N]$ anyways. The TPs of these sequences are, respectively, low and high in English, and both non-zero in Hindi. Sensitivity to N in $[\tilde{V}N]$ should thus be average in English; in Hindi, sensitivity to N in $[\tilde{V}N]$ may be high (if it is a low-probability oddball) or average (if it is high-probability). Sensitivity to C in $[\tilde{V}C]$ should also be predictable from its probability. The fact that Hindi successfully contrasts $[\tilde{V}N]$ and $[\tilde{V}C]$ may not be about sensitivity to N so much as sensitivity to C after $[\tilde{V}]$.

In languages such as Ikalanga, the probability of $\tilde{V}$ is high after N , and low after NC , but if these two environments cannot be distinguished, then the prob-

ability of $\tilde{V}$ after N/NC may be high, middling, or low, depending on the relative commonness of N and NC. The probability of V after N/NC would also depend on the relative commonness of N and NC. If one of V and $\tilde{V}$ was high probability, and the other low-probability, then presumably, the low-probability vowel would be an oddball and sensitivity to it is predicted to be high. Alternatively, if they are equally common, sensitivity to both would be average.

To sum up, the selective adaptation account predicts generally high sensitivity to $\tilde{V}$ after C, and high or average sensitivity to $\tilde{V}$ after N, depending on the contrasts present in the language. Notably, it does not predict low sensitivity to $\tilde{V}$ in any context, because the confluence of low probability and high similarity to high-probability sounds which results in low sensitivity does not occur.

5.3 Interpreting the typological effects of habituation and selective adaptation

One of the motivations for this paper's existence is an apparent typological asymmetry attributable to the asymmetric perception of carryover and anticipatory nasal coarticulation. In this paper, I have discussed some of the typological predictions made by the perceptual wink and selective adaptation models of auditory bias and language-specific knowledge.

I must note that these typological predictions are psychophysical rather than phonological in nature. This is because formal phonology and psychophysical models are situated on different levels of analysis in the sense of Marr (1982). Specifically, the psychophysical models are algorithmic: they make claims about mental processes which deal with low-level information like formant bandwidth or probability. Phonology is computational in the sense that it makes claims about a formal system which deals with highly abstract units of information, such as nasality or syllable role. If phonology is a description of the principles of a computer's operating system, then psychophysics is a description of the binary logic used in the computer's core processor.

Strictly speaking, the psychophysical models do not rule out any phonological computations; they should not be taken as evidence for, e.g., a set of Optimality Theoretic constraints. In terms of Marr's (1982) levels of analysis, phonological theories are descriptions of the computational level (what the system accomplishes, or its goal), while psychophysical models are descriptions of the algorithmic level (how the system carries out its work, or the particular method by which the system reaches its goal). Any constraint imposed on the phonology by the psychophysical processes discussed here will take the form of limitations on the general architecture of the computational system, rather than constraints or rules modeled as objects within that system.

In this discussion, I have limited myself to speculating about some predicted typological effects of habituation and selective adaptation while remaining agnostic about the details of the relationship between psychophysics and phonology beyond what was stated in the previous paragraph. Attempting to model these effects without a theory of their origin in that relationship results in approaches which confuse the computational and algorithmic levels of analysis by

attempting to place psychophysical constraints directly in the phonology.

For example, Stanton (2016) observes that the presence of /N NC/ and /C NC/ contrasts in contexts where the cues to the contrast are poor (e.g., /N NC/ in word-final position) entails their presence in contexts where the cues are rich (e.g., /N NC/ in intervocalic position). To account for this generalization, Stanton proposes a set of OT constraints which penalize /N NC/ and /C NC/ contrasts which are insufficiently perceptually distinct, as well as a constraint which penalizes the neutralization of these contrasts. While this OT analysis correctly generates the observed typology with regards to the distribution of /N NC/ and /C NC/ contrasts, it is ultimately a descriptive restatement of a set of generalizations for which the explanatory mechanism may be auditory in nature. Minimally, a theory of how perceptual distinctiveness is determined and how this information is passed to the phonology is necessary.

Approaches which attempt to model psychophysical effects in the phonology can only ever be descriptive rather than explanatory. What is needed instead is a theory of how psychophysics constrains phonology without being embodied as formal constraints. This topic requires much deeper investigation than is possible here, and I leave it as an avenue for future investigation.

6 Conclusion

The goal of the research program set out in this paper is to determine why carryover and anticipatory nasal coarticulation are perceived asymmetrically. Two experiments were conducted towards this end. Experiment 1, a categorization task with gated stimuli, showed that English listeners can use anticipatory but not carryover nasal coarticulation as evidence for the presence of adjacent nasal consonants. Experiment 2, a 4IAX discrimination task, showed that when the influence of language-specific knowledge is reduced, there ceases to be asymmetric perception in favor of anticipatory nasal coarticulation.

Two psychophysical models, the perceptual wink model of activation and habituation and a model of selective adaptation, were discussed as specific models of the auditory bias and language-specific knowledge hypotheses, respectively. The results of experiments 1 and 2 were used to evaluate these models, with the conclusion that each is compatible with certain aspects of the results, but that the experiments provided no evidence which could be used to rule out either one.

One shortcoming of the experiments conducted so far is that almost all participants were native listeners of English. Both experiments should be rerun with listeners of languages which i) have different patterns of nasal coarticulation, and/or ii) give nasalized vowels different phonological status, to determine how these interact with the perception of carryover and anticipatory nasal coarticulation. For example, Spanish is argued to have phonetic but not phonological (i.e., mechanical but not targeted) nasal coarticulation (Solé 1995); French contrasts oral and nasal vowels and has both carryover and anticipatory nasal coarticulation, though their details differ from those of English (Delvaux et al.

2008) (though see Pouplier et al. (2023) for an argument that the phonological status of nasal vowels doesn't predict coarticulatory patterns); and Brazilian Portuguese has both phonemic and allophonic nasal vowels, though their articulations differ somewhat (Marques & Scarborough 2017). Any of these languages would be useful points of comparison with English.

Another potential problem (but also opportunity) is that individuals' patterns of coarticulation may differ considerably. Beddor et al. (2018) and Zellou (2022) argue that an individual's patterns of coarticulation in production (partially) predict how they perceive coarticulation; an additional experiment might attempt to replicate and expand on these findings, e.g., extending them to carryover nasal coarticulation.

The results of Zellou (2022) also suggest another avenue for future investigation. One critical finding was not only that the pattern of nasal coarticulation varies greatly between speakers, but that data must be gathered from many participants in order to detect such variation; Zellou's Experiment 1 involved sixty participants. Many of the studies which established the pattern of nasal coarticulation in other languages had relatively few participants. For example, Solé (1995) argued that anticipatory nasal coarticulation in Spanish is "mechanical" (not linked to the duration of the vowel, as in English) on the basis of data from just three Spanish speaking participants. It is possible that individual variation similar to that discovered by Zellou exists in other languages. Continuing with Spanish as our example, we could imagine that some speakers have the "mechanical" pattern of anticipatory nasal coarticulation discussed by Solé, while others might have extensive coarticulation (and perhaps perceptual attunement to that coarticulation) similar to that found in English, perhaps especially if they are Spanish/English bilinguals—though this is speculative. A study similar to Zellou's Experiment 1 involving Spanish speaking participants may be in order.

A great deal of work remains for the future. So far, two avenues of attack on the problem of asymmetric perception of nasal coarticulation have been pursued: Experiment 1 replicated some previous findings about the perception of nasal coarticulation and demonstrated the need to test each hypothesis separately, and Experiment 2 attempted to falsify the auditory bias hypothesis. But an additional approach, attempting to support the language-specific knowledge hypothesis, was pursued only indirectly via arguments that the falsification of the auditory bias hypothesis constitutes indirect support for the language-specific knowledge hypothesis. Further work is needed to directly support this hypothesis.

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References

- Beddor, Patrice Speeter, Andries W. Coetzee, Will Styler, Kevin B. McGowan & Julie E. Boland. 2018. The time course of individuals’ perception of coarticulatory information is linked to their production: implications for sound change. *Language* 94(4). 931–968.
- Beddor, Patrice Speeter & Chutamanee Onsuwan. 2003. Perception of prenasalized stops. In *Proceedings of the 15th international congress of phonetic sciences (ICPhS)*, vol. 1, 407–410.
- Boersma, Paul & David Weenink. 2023. *Praat: doing phonetics by computer*. computer program. Version 6.4.01. <http://www.praat.org/> (10 December, 2023).
- Chen, Marilyn Y. 1997. Acoustic correlates of english and french nasalized vowels. *The Journal of the Acoustical Society of America* 102(4). 2360–2370.
- Cohn, Abigail C. 1993. Nasalisation in English: phonology or phonetics. *Phonology* 10. 43–81.
- Cutler, Anne. 2012. *Native listening: Language experience and the recognition of spoken words*. 1st edn. Cambridge, Massachusetts: MIT Press. <https://doi.org/10.7551/mitpress/9012.001.0001>.
- Delvaux, Véronique, Didier Demolin, Bernard Harmegnies & Alain Soquet. 2008. The aerodynamics of nasalization in French. *Journal of Phonetics* 36(4). 578–606.
- Diehl, Randy L, Andrew J Lotto & Lori L Holt. 2004. Speech perception. *Annu. Rev. Psychol.* 55. 149–179.
- Fowler, Carol A. & Julie M. Brown. 2000. Perceptual parsing of acoustic consequences of velum lowering from information for vowels. *Perception & Psychophysics* 62. 21–32.
- Gerrits, Ellen & Martin E. H. Schouten. 2004. Categorical perception depends on the discrimination task. *Perception & Psychophysics* 66(3). 363–376.
- Holt, Lori L & Andrew J Lotto. 2010. Speech perception as categorization. *Attention, Perception, & Psychophysics* 72(5). 1218–1227.
- Huber, David E & Randall C O’Reilly. 2003. Persistence and accommodation in short-term priming and other perceptual paradigms: temporal segregation through synaptic depression. *Cognitive science* 27(3). 403–430.

- Jeong, Sunwoo. 2012. Directional asymmetry in nasalization: a perceptual account. *음성음운형태론연구 (Studies in Phonetics, Phonology, and Morphology)* 18(3). 437–469.
- Keating, Patricia A, Michael J Mikoś & William F Ganong. 1981. A cross-language study of range of voice onset time in the perception of initial stop voicing. *The Journal of the Acoustical Society of America* 70(5). 1261–1271.
- Klatt, Dennis H. 1976. Linguistic uses of segmental duration in english: acoustic and perceptual evidence. *The Journal of the Acoustical Society of America* 59(5). 1208–1221.
- Krakov, Rena A. 1999. Physiological organization of syllables: a review. *Journal of Phonetics* 27(1). 23–54.
- Ladefoged, Peter & Ian Maddieson. 1996. *The sounds of the world's languages*. Oxford, UK & Cambridge, Massachusetts: Blackwell Publishers.
- Lahiri, Aditi & William Marslen-Wilson. 1991. The mental representation of lexical form: a phonological approach to the recognition lexicon. *Cognition* 38(3). 245–294.
- Marques, Luciana & Rebecca Scarborough. 2017. Perception and acoustics of vowel nasality in Brazilian Portuguese. In *Interspeech*, 616–620.
- Marr, David. 1982. *Vision*. The MIT Press.
- Norris, Dennis, James M McQueen & Anne Cutler. 2003. Perceptual learning in speech. *Cognitive psychology* 47(2). 204–238.
- Ohala, John J, Carrie S Masek, Roberta A Hendrick & Mary Frances Miller. 1981. The listener as a source of sound change. *Papers from the Parasession on Language and Behavior*.
- Ohala, John J. & Manjari Ohala. 1995. Speech perception and lexical representation: the role of vowel nasalization in hindi and english. In Bruce Connell & Amalia Arvanti (eds.), *Papers in laboratory phonology IV: phonology and phonetic evidence*, 41–60. Cambridge, United Kingdom: Cambridge University Press.
- Parker, Ellen M, Randy L Diehl & Keith R Kluender. 1986. Trading relations in speech and nonspeech. *Perception & Psychophysics* 39(2). 129–142.
- Peirce, Johnathan, Jeremy R. Gray, Sol Simpson, Michael Macaskill, Richard Höchenberger, Hiroyuki Sogo, Erik Kastman & Jonas Kristoffer Lindeløv. 2019. Psychopy2: experiments in behavior made easy. *Behavior Research Methods* 51. 195–203. <https://doi.org/10.3758/s13428-018-01193-y>.
- Pisoni, David B. 1973. Auditory and phonetic memory codes in the discrimination of consonants and vowels. *Perception & psychophysics* 13. 253–260.
- Pisoni, David B. & Paul A. Luce. 1987. Trading relations, acoustic cue integration, and context effects in speech perception. In Marten Egbertus Hendrik Schouten (ed.), *The Psychophysics of Speech Perception*, 155–172. Dordrecht, Netherlands: Martinus Nijhoff Publishers. <https://doi.org/10.1007/978-94-009-3629-4>.
- Poupplier, Marianne, Francesco Rodriquez, Roy Alderton, Justin J. H. Lo, Eva Reinisch, Bronwen Evans & Christopher Carignan. 2023. The window of opportunity: anticipatory nasal coarticulation in three languages. In Radek Skarnitzl & Jan Volín (eds.), *Proceedings of the 20th international congress*

- of phonetic sciences (ICPhS), 2085–2089. Prague: Guarant International. <https://www.icphs2023.org/programme/proceedings/>.
- Repp, Bruno H. 1979. Relative amplitude of aspiration noise as a voicing cue for syllable-initial stop consonants. *Language and Speech* 22(2). 173–189.
- Repp, Bruno H. 1982. Phonetic trading relations and context effects: new experimental evidence for a speech mode of perception. *Psychological bulletin* 92(1). 81.
- Rusconi, Patrice & David E. Huber. 2018. The perceptual wink model of non-switching attentional blink tasks. *Psychonomic Bulletin & Review* 25. 1717–1739.
- Scarborough, Rebecca & Georgia Zellou. 2013. Clarity in communication: “clear” speech authenticity and lexical neighborhood density effects in speech production and perception. *The Journal of the Acoustical Society of America* 134(5). 3793–3807.
- Scarborough, Rebecca, Georgia Zellou, Armik Mirzayan & David S. Rood. 2015. Phonetic and phonological patterns of nasality in lakota vowels. *Journal of the International Phonetic Association* 45(3). 289–309.
- Simpson, Andrew J.R., Nicol S. Harper, Joshua D. Reiss & David McAlpine. 2014. Selective adaptation to “oddball” sounds by the human auditory system. *Journal of Neuroscience* 34(5). 1963–1969.
- Solé, Maria-Josep. 1995. Spatio-temporal patterns of velopharyngeal action in phonetic and phonological nasalization. *Language and Speech* 38(1). 1–23.
- Stanton, Juliet. 2016. Predicting distributional restrictions on prenasalized stops. *Natural Language & Linguistic Theory* 34. 1089–1133.
- Styler, Will. 2015. *On the acoustical and perceptual features of vowel nasality*. University of Colorado dissertation.
- Styler, Will. 2017a. *Nasality Automeasure Script Package*. Version 5.9. https://github.com/stylerw/styler_praat_scripts/tree/master/nasality_automeasure (10 December, 2023).
- Styler, Will. 2017b. On the acoustical features of vowel nasality in english and french. *The Journal of the Acoustical Society of America* 142(4). 2469–2482. <https://doi.org/10.1121/1.5008854>.
- Zehr, Jérémy & Florian Schwarz. 2018. Penncontroller for internet based experiments (ibex). <https://doi.org/10.17605/OSF.IO/MD832>.
- Zellou, Georgia. 2022. Coarticulation in phonology. In *Cambridge Elements in Phonology*. Cambridge University Press. <https://doi.org/10.1017/9781009082488>.